Velocity graphs and components

LEARNING TARGET · R2 I can read v–t areas and resolve velocity into components.

A · KEY VOCABULARY

Signed area	Above zero adds displacement; below zero subtracts it.
Distance	Add area sizes, including the return trip.
Component	A vector's signed part along one axis.
Angle θ	Name the starting axis and turn direction.

B · KEY EQUATIONS + UNITS

$$\Delta x = \text{signed area}$$

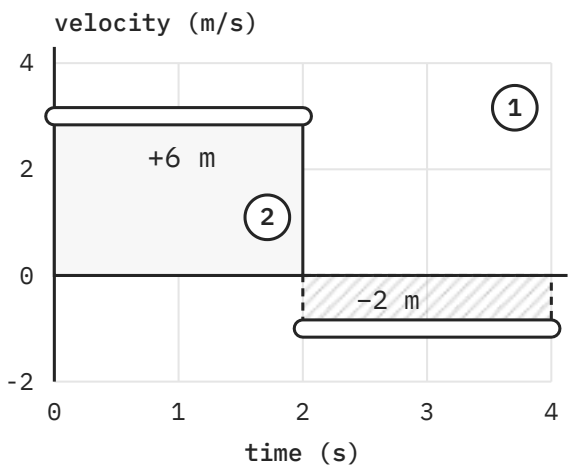
Split v along fixed axes

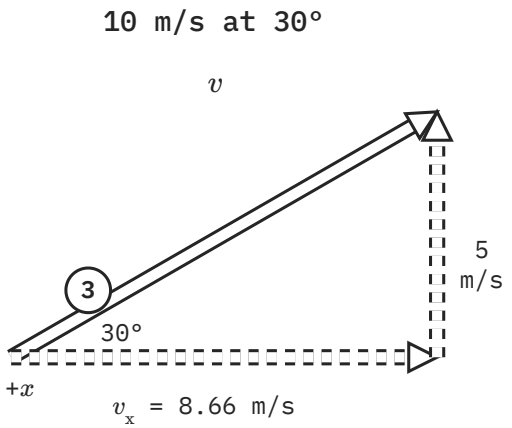
v_x $v \cos\theta$

v_y $v \sin\theta$; θ from $+x$

C · COLOR IT · AREA AND COMPONENTS

Two views of motion.





☐ **1 · PURPLE** Velocity and its parts

☐ **2 · BLUE** Signed area: displacement

☐ **3 · PENCIL** Axes and angle

D · WORKED EXAMPLE

A cart moves at **+3 m/s** for **2 s**, then **−1 m/s** for **2 s**. **Find its displacement and total distance.**

GIVEN	EQUATION	WORK	ANSWER
v_1 = **+3 m/s** v_2 = **−1 m/s** **2 s** per interval	Use areas because each interval has steady v.	Δx = 3(2) − 1(2) distance = 6 + 2	**+4 m; 8 m** Return travel reduces Δx, not distance.

E · THE 2026-XJ THREAD

DAY 4 OF 22 · NEW TOOL: SIGNED AREA

For fictional XJ, the area of a velocity–time graph gives a change in position. Components separate motion toward Earth from motion across its line of sight.

Unit question: How fast is it moving, and where will it arrive?

UNIT 1 COLOR CODE ☐ green speeding up ☐ red slowing down ☐ yellow steady · balanced ☐ blue position · displacement ☐ purple velocity ☐ orange forces pencil · everything else